PIC: Status, Pending Work and the Production Plan

AlphaHelion Procurement Intelligence Cloud. Status as of 28 Oct 2026, written after the module build (agents A–E) and the dataset build (six dataset engineers).

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1. Summary

Area	State
Core workflows F1–F9	**Implemented**, including every gateway branch, which are now exercised by dataset D1 (cases 9–34)
Operations modules (amendments, auctions, EDI, payment files, e-invoice, 4-way match, reconciliation, MDM-04, punch-out, ERP import, notifications, MFA, SSO)	**Implemented**, exercised by dataset `module_cases`
Purchase types (expense, capital asset, consignment, subcontracting)	**Implemented in this phase**, exercised by D2
Industry packs (construction, IT services, logistics, healthcare, general)	**Implemented** as five demo tenants, D3–D7
Production readiness (security headers, rate limits, health/readiness, metrics, privacy, CI, container files, K8s/Terraform, runbooks)	**Implemented** (agent E)
Datasets	**D1–D9 and D11 done** (each engineer's full suite green: 230–249 tests); D8 generator done, full-scale runs need a production-like environment; D10 needs your ERP extract
Code size	249 backend tests, 244 API routes, 114 tables, 44 screens, migrations 0001–0005
Your decisions (28 Oct)	**Vercel** for now (GCP for the market later); **demo datasets on the production Neon database**; Neon **Singapore** is fine (demo data only)
Biggest risk right now	**About 140 changed files are not committed or pushed.** Nothing from the module and dataset phases is on GitHub or Neon yet.

The product is functionally complete against the blueprint for an India-first manufacturing tenant and the five industry packs. What separates it from a production go-live is:

- **Integration of this phase:** one full test run with every pack, then commit, push and migrate Neon.
- **A short list of code hardening items** (§5.1).
- **External items only you can provide:** hosting, credentials, legal and tax sign-off.

2. What is in place

Layer	Contents
Backend	FastAPI modular monolith, schema `pic`, forced row-level security per tenant, append-only hash-chained audit, transactional outbox, workflow engine with timers and SoD, effective-dated rules with scopes, Idempotency-Key, problem+json errors
Integration layer	Connector registry with per-company ERP target (rule `ERP-TARGET`). SAP S/4 is a **simulator**; the EDI, SMTP, bank SFTP, GST IRP, OCR, e-sign and sanctions adapters are ready but need credentials
Frontend	Next.js workspace with 44 screens, a supplier portal, a BFF proxy with CSP/HSTS, multipart uploads and trace propagation
Operations	`/healthz`, `/readyz`, `/metrics` (golden signals, business SLOs, outbox lag), JSON logs with redaction, optional OpenTelemetry, worker heartbeat
Security	Rate limits per IP and per user, body-size limits, MFA (TOTP) with admin reset, OIDC SSO, field encryption, privacy export and erasure (DPDP)
Delivery	GitHub CI (Python 3.14 tests, migrations, pip-audit gate, npm audit, tsc/eslint, container builds), security scans (gitleaks, CodeQL, ZAP report-only), release to GHCR, Neon deploy workflow, `deploy/k8s` + `deploy/terraform` (GKE Autopilot, asia-south1), 10 runbooks
Data	Axion signature case (Oct 2026 – Mar 2027), migrated legacy tenants GMC and DBY, dataset packs on top of them (§4)

3. Workflows: implementation status

3.1 Core workflows F1–F9

Flow	Built	Branches exercised by data	Still open
F1 MRP → planned order → requisition	Yes	Simulation run, PAST_DUE, RESCHEDULE_IN/OUT, CANCEL, firming, quota sourcing, ERP import (accepted/rejected rows)	**Performance**: 50k materials takes 2–4 h against a 10 min target
F2 Guided buying → requisition	Yes	Return, reject, resubmit, value change re-approval, budget soft/hard, category budget warning/hard block, punch-out	Maintenance screens for budgets (loaded directly today)
F3 Sourcing → award	Yes	RFQ, reverse auction with soft-close, clarification board, rejected strategy recommendation, award without approval	**Sourcing event cancellation**; **award-approval rejection** path
F4 Contract lifecycle	Yes	Below-₹1 Cr rejection, redline, 80/90/100% alerts, renewal approved/rejected, expiry, variation, extension, termination, index reset	E-sign needs a provider (internal signatures today)
F5 PO → confirmation → delivery	Yes	Supplier rejection, cancellation, confirmation breach, change in/out of tolerance, EDI ORDRSP/ORDCHG, schedules, **cancel/close (new)**	**E-mail/EDI cancellation notice** to the supplier; screen buttons for cancel/close
F6 GR / SES → quality	Yes	Wrong-PO reversal, rejected lot → return → CAPA, rejected SES, lot/expiry/serial checks, consignment, subcontracting	**Validation that a GR's ASN belongs to the same PO**
F7 Invoice → match → post → pay	Yes	2/3/4-way, ERS, dispute, IRN mismatch, ERP timeout → dead letter → replay, credit memo, retention, freight audit, payment-file rejection	**Payment-run selection by supplier/invoice** (runs are tenant-wide today); capitalisation in the GR ERP payload
F8 Supplier onboarding	Yes	Invitation, self-registration, qualification rejected, certificate expiry → Conditional, bank change rejected, MDM-04 address/contact/tax	Real sanctions feed
F9 Material governance	Yes	Steward rejection, duplicate warning band, block + override	**GST rate on a material request** (every new material defaults to 18%)

3.2 Intelligence and governance flows

Flow	State
Spend classification → steward review	Built. D9 result: 0.991 overall; the curated set scores only 0.733 because keywords match inside words ("ecu" in "security"). Fix planned (§5.1 #13)
Supplier de-duplication	Built. D9: registration screening precision 0.78 / recall 0.75 (no PAN or "Pvt Ltd" normalisation); spend clusters 0.83 / 1.0
Category plan → budget check	**Done in this phase** (hard and warning rules on requisitions)
Demand sensing	Built. D9: MAPE 0.037 at horizon 1 (ERP plan 0.061); spend forecast MAPE 0.035

Flow	State
Copilot	Built. D9: answer-contract completeness 1.0 (gate ≥ 0.99 met) on 40 questions
Risk signal → mitigation	Built. Needs a live feed
CAPA	Open, respond, verify and close all exercised (D1 case 28)
Reconciliation	Difference → resolution → clean rerun (module_cases REC-102/103)
Agent autonomy suspension (< 70% acceptance)	Built. Needs ≥ 20 feedback events from live use
Approval-matrix simulation (ADM-03)	**Done** (Admin → Approval simulation)

3.3 Module behaviour added or fixed in this phase

The datasets exposed these gaps; each one is fixed and covered by tests.

Area	Change
Payables	Retention withholding and release (construction); freight audit (logistics); ERS settlement for consignment; credit memo to the ERP after posting; dispute keeps the exception open
Orders	PO cancel and close; reversed-receipt state; subcontracting bill of materials; catalogue lines take HSN and GST from the catalogue item (they were hard-coded to 18%)
Receiving	Consignment stock, component provision and consumption (541/543/544), batch/expiry/serial checks, lot trace and recall, full GR reversal (inspection lot, ASN, ERP 102)
Requisitions	Asset lines need an asset or WBS; consignment and subcontracting lines need a material; a value change re-books the budget
Contracts	Late extension re-dated; cancelled POs skipped in consumption; index reset from the latest negotiated price
Suppliers	**Certificates never expired** (the timer fired on the expiry day itself): fixed
Planning	RESCHEDULE_OUT and CANCEL exceptions
Templates	General: goods receipt above ₹1 L; Logistics: carriers need insurance and a permit
Platform	Per-company ERP target; MFA admin reset; supplier portal company tab; OCR invoice upload; binary uploads through the proxy
Tax	New GST check: an invoice whose rate is not a scheduled rate for its HSN/SAC raises a TAX exception (CBIC notifications in force 22.09.2025)
Admin	Plant holiday calendars maintained through a service, with source and audit

4. Datasets: status

#	Dataset	Status	Tenant / window	Content
D1	Branch cases	**Done** (25 tests)	AXN, Apr 2027	Cases 9–34, every F1–F9 gateway outcome
D1b	Operations module cases	**Done** (8 tests)	AXN, May 2027	Amendments, auction, EDI, e-way bill, 4-way match, reconciliation, payment files, MDM-04, punch-out, ERP import
D2	Purchase types	**Done** (8 tests)	AXN, Jun 2027	Expense, capital asset with capex committee and WBS hard stop, consignment + ERS, subcontracting
D3	Construction	**Done**	CON	SC-118, WBS C-2207, running-account bills at 35/70/100%, 5% retention, VO-01, release
D4	IT services	**Done**	ITS	SOW-4410, RC-22, 320 h timesheet, licence renewal alert and extension
D5	Logistics	**Done** (with D7: 12 tests)	LOG	Lane CHE→PNQ ₹54/km with a demo fuel index, 15 trips, proof of delivery, freight audit and credit note
D6	Healthcare	**Done** (8 tests)	HLT	KN-77 consignment implants, lot/expiry/serial, usage-triggered replenishment, recall trace
D7	General enterprise	**Done**	GEN	Laptop punch-out ₹78,500 with one approval, catalogue, housekeeping service
D8	Volume	**Generator done** (3 tests)	Local / staging only	50k materials, invoices, spend lines. The real runs need a production-like environment
D9	AI evaluation sets	**Done** (7 tests)	AXN	2,568 + 86 labelled spend lines, 18 forecast actuals, 48 duplicate pairs, 40 Copilot questions; evaluator `python -m app.pic.tools.ai_eval`
D10	Real tenant masters	**Needs your ERP extract**	Your tenant	Vendors, materials, contracts, open POs, open AP, 24–36 months of history
D11	Reference data	**Done** (9 tests)	All tenants	370 HSN/SAC rates from CBIC notifications, 212 holidays (central + 7 states 2026, central + TN 2027), 56 UNSPSC mappings, FBIL FX rates of 28 Oct 2026

Still to do on the datasets:

- **One combined run:** a full-suite run with every pack loaded in one database. Each engineer ran in their own database.
- **Order-dependent budgets:** next year's (FY 2027-28) AXN budgets are created by three packs with different amounts, so whichever pack runs first wins. They will be defined once.
- **One AXN payment run pays a signature-case invoice** (INV-TF-9031). Fixed by adding supplier selection to payment runs.
- **GSTIN check character:** the Axion company GSTIN `33AAACA4321F1Z9` fails the check. It becomes `33AAACA4321F1ZF` in the code and as an audited correction on Neon.
- **Duplicate helpers:** the GSTIN and tenant helpers exist in four copies across the packs. They will be merged into `_tenant_kit`.
- **Pack order:** reference data runs first, so it doesn't reach the five industry tenants created later. It will run last as well.
- **Invalid seed GSTINs** in the signature case (e.g. `33AABCM7890P1Z4`), and supplier registration doesn't validate the check character. Both are fixed together.

5. Pending work

5.1 Code: can be done by me, no external input

#	Item	Why	Size
1	Integrate all agent changes, then one full test run on one database	Payables, orders and receiving were edited by several engineers	2–3 h
2	Commit and push; CI green on GitHub	Nothing of this phase is saved remotely	30 min
3	Payment-run selection by supplier, invoice or due date	Required for real AP; fixes the dataset side effect	1 h
4	Single FY 2027-28 budget definition; GSTIN fix; merge the helpers	Data correctness	1 h
5	Replace `python-jose` with PyJWT	Critical advisory with no fix released; not exploitable today, but it fails an audit	1 h
6	Token revocation (token version per user, migration 0006)	A deactivated or erased user keeps a valid token for up to 8 h	1.5 h
7	MRP quick wins: composite indexes (migration 0006), cached source determination, bulk inserts	50k materials is 2–4 h today; quick wins should bring it to about 20–40 min, but the 10 min target needs a deeper rewrite	3–4 h
8	Screens: PO cancel/close, consignment withdrawal and settlement, lot trace and recall, subcontracting components	The APIs exist but have no buttons	3 h
9	Hide the "Legacy demo screens" link in production; move `PIC_ESIGN_*` into `config.py`	Production surface	30 min
10	GR ASN-to-PO check; EDI/e-mail cancellation notice; sourcing event cancel; award-approval rejection	Spec gaps found by the datasets	3 h
11	Docs (gap analysis, remaining-work report, process flows, README, datasets) and PDFs	Keep the documents true	1.5 h
12	Apply 0004–0006, the rules and all packs to the production Neon database (your decision); verify	Neon still has only 0001–0003	1 h
13	Classifier word-boundary matching; PAN and suffix normalisation in registration de-duplication	D9 findings	1.5 h
14	**Vercel deployment** (§6.1): API entrypoint and `vercel.json`, a worker tick through Vercel Cron plus a drain after each write, Neon pooled connection, web project settings, deployment guide	Your hosting decision	3 h

5.2 Code that needs a schema decision, later

Item	Proposal
GST rate on a material request	Add `material_request.gst_rate`

Item	Proposal
HSN/SAC on free-text lines	Add the columns to requisition and PO lines
Job-work tax code on subcontracting lines	A separate tax code per line (SAC 9988)
Retention as its own items	A `retention_item` table instead of release documents as invoices
Consumption origin and usage numbers	Add origin `CONSUMPTION` and a usage-document number range
Weight on a service entry	For cost per tonne-km
Field-encryption key version	Needed to rotate `PIC_FIELD_KEY`
Maintenance screens for budgets, catalogue, taxonomy, planning data, demand, order channel	Loaded directly today, with audit
Other features	WBS re-check on PO variations, advance recovery, accessorials, capitalisation payload, clinical sign-off on awards

5.3 Cannot be finished in code alone

- **Live integrations:** real ERP posting, GST IRP, e-sign, bank host-to-host, EDI, sanctions, SMTP, OCR and the LLM gateway all need credentials (§8).
- **Performance runs:** k6 latency (P95 < 400 ms), cube < 3 s and the 10,000-user test need a staging environment.
- **Docker:** a local image build needs Docker Desktop repaired. GitHub CI builds the images in any case.
- **Reviews and decisions:** legal retention periods, the tax team's rate confirmations, a penetration test and a ZAP triage.

6. What "production grade by tomorrow" can and cannot mean

Achievable by tomorrow: a production release candidate.

- All code integrated and tested in one run, committed, CI green, and container images built by CI.
- Neon migrated with reference data, and the hardening items §5.1 #3–#9 done.
- Integrations whose credentials are missing switched off cleanly, rather than half-configured. They surface as "needs configuration" in the admin connector screen.
- That is a system you can put real users on for a **pilot**: requisitions, approvals, sourcing, contracts, POs, receiving, invoice capture and matching, with ERP posting through the simulator or a manual export.

Not achievable by tomorrow, whoever does the work:

- Live money and tax flows (bank files to a real bank, IRN from the GST portal, ERP postings) until you provide credentials and the tax team signs off.
- The MRP 10-minute target at 50,000 materials.
- Proven latency and availability SLOs (needs staging and load tests) and an external penetration test.
- India data residency, if required: Neon is in Singapore today.

Your decision on demo data: the demo datasets go into the production Neon database, because there are no real customers yet. Before the first real customer is onboarded, plan a split (a Neon branch for demo, or removing the demo tenants).

6.1 Deployment on Vercel (now) and GCP (for the market)

Part	On Vercel	Notes
Web (Next.js)	Vercel project with root `frontend`	Native. `PIC_API_URL` points at the API project; `PIC_ENV=production`
API (FastAPI)	Vercel project with root `backend`, Python **3.14** (supported), entrypoint `app/main.py:app`	Zero-config FastAPI; function region **sin1 (Singapore)**, next to Neon
Worker (outbox, timers, connectors)	No long-running process on Vercel. Instead: **Vercel Cron** calls a protected worker-tick endpoint, and each write request drains its own tenant's outbox after the response	**Vercel Pro** allows a tick every minute; **Hobby allows only one cron a day**, so timers (SLA escalations, certificate expiry) would wait until the next request or the daily run
Database	Neon, **pooled** host for the API, direct host for migrations	Tenant settings are transaction-local (`SET LOCAL`, `set_config(…, true)`), so they are safe with the pooler
Live updates (`/v1/stream`)	Streaming works within the function time limit; the browser reconnects	Acceptable for a demo
Rate limits, login lockout	In-memory per instance, so weaker on serverless	Add Vercel Firewall rules; a shared store later on GCP

Part	On Vercel	Notes
Migrations and datasets	GitHub `neon.yml` (unchanged)	Runs on each push to `main`

GCP (`deploy/k8s` , `deploy/terraform` , GKE Autopilot asia-south1) stays the target for the market. It runs the worker as a real process and removes the serverless limits above.

7. The plan to tomorrow

When	Work	Done when
Tonight, block 1 (≈ 4 h)	Last dataset agent reports → items 1, 3, 4 → full suite on one database with every pack → tsc + eslint → commit + push	Full suite green; CI green on GitHub
Tomorrow morning, block 2 (≈ 4 h)	Items 5, 6, 7 (migration 0006), 9 → full suite again	Security items closed; MRP measured again and the new number reported
Tomorrow midday, block 3 (≈ 3 h)	Items 8, 10, 13 → browser check of every screen on localhost → item 11 docs + PDFs	Screens and docs match the code
Tomorrow afternoon, block 4 (≈ 4 h)	Item 14 Vercel deployment → item 12 Neon (production database, all packs) → `/readyz` and verification report → browser check on the Vercel URL → go/no-go (§9)	Live on Vercel; release tagged

The plan assumes no new large scope. Anything found in the combined test run is fixed first, ahead of blocks 2–4.

8. What you must supply or decide

Decisions (needed before tomorrow afternoon)

#	Decision	Default if you don't decide
D-1	Hosting	**Decided:** Vercel now, GCP for the market
D-1a	**Vercel plan: Pro or Hobby?**	Pro is needed for a worker tick every minute; on Hobby, background work runs only after requests and once a day
D-2	Demo data on production Neon	**Decided:** same database (demo only, no real customers yet)
D-3	Data residency	**Decided:** Neon Singapore for now; Vercel functions in sin1
D-4	**Identity**: company SSO through OIDC, or built-in passwords + MFA	Built-in passwords + MFA
D-5	**Neon plan**: history retention ≥ 7 days, daily export	Current plan; RPO not guaranteed

Supplies

Item	Needed for
Rotate the Neon password (it was shared in chat earlier) and update `backend/.env` and the GitHub secrets	Security, before go-live
GitHub secrets: `NEON_DATABASE_URL`, `PIC_SEED_PASSWORD`, `PIC_JWT_SECRET` (≥ 48 chars), `PIC_FIELD_KEY`, `PIC_METRICS_TOKEN`, `GITLEAKS_LICENSE` (organisation repos)	CI, release, Neon deploy
Vercel: access to your team (`vercel login` on this machine, or a `VERCEL_TOKEN` + team/org ID as GitHub secrets) and the plan (Pro/Hobby)	Deploying
Domain name (optional; Vercel gives `*.vercel.app` with TLS)	Go-live
ERP sandbox or extract (SAP S/4 first): vendors, materials, contracts, open POs, open AP, history (D10)	Real data, replacing the simulator
Credentials: SMTP, EDI gateway, bank SFTP/host-to-host + the real debtor account, GST IRP/GSP, OCR, e-sign, sanctions feed, OIDC, LLM gateway	Live integrations
Tax team: confirm GST for goods transport (SAC 996791 at 18%), KN-77 at 5%, ERS (self-billing) under GST, job-work SAC 9988	Correct tax
Reference: FX history (only 28 Oct 2026 is loaded; the FBIL archive is behind a captcha), 2027 holiday lists for Karnataka, Maharashtra, Telangana, Gujarat, Delhi and Haryana (not published yet), real IFSC codes, a real diesel index feed, the licensed UNSPSC code set if class level is needed	Reference data
Legal: retention periods per data category, lawful-basis wording	Privacy
Business: FY 2027-28 budgets (the demo copies FY 2026-27), the NPPA ceiling check for implant prices	Demo realism

Item	Needed for
Docker Desktop repair (steps given earlier)	Local image builds only

9. Go / no-go checklist

- [ ] Full backend suite green in one database with every pack; tsc and eslint clean
- [ ] Everything committed and pushed; CI, security and release workflows green
- [ ] Migrations 0001–0006 on Neon; `alembic check` clean; RLS on every table; audit chain verifies
- [ ] `PIC_ENV=production`; legacy demo routers off; API docs off; CORS empty; production secrets set
- [ ] Neon password rotated; no secrets in the repository (gitleaks clean)
- [ ] Integrations without credentials are switched off and shown as "needs configuration"
- [ ] Browser check of every screen as buyer, approver, AP clerk, supplier and admin
- [ ] Decisions D-1 to D-5 recorded
- [ ] Known limits accepted in writing: MRP scale, no load test yet, Singapore residency (if kept), simulator ERP